

Demo Answer Sheet

Subject: Mathematics

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Q1. Solve for x: $2x + 5 = 15$

$$2x + 5 = 15$$

$$2x = 10$$

$$x = 5$$

Q2. What is the derivative of $x^2 + 3x$?

$$d/dx (x^2 + 3x) = 2x + 3$$

Q3. Find the area of a circle with radius 7cm.

$$\text{Area} = \pi * r^2 = 3.14159 * 49 = 153.94 \text{ cm}^2$$

Q4. Simplify: $(3x^2)(2x^3)$

$$= 6x^5$$

Q5. What is the value of $\sin(30 \text{ degrees})$?

$$\sin(30) = 0.5$$

Demo Answer Sheet (continued)

Q6. Solve the equation: $x^2 - 4 = 0$

$$x^2 = 4$$

$$x = \pm 2$$

Q7. What is the integral of $2x$?

$$\text{Integral of } 2x \, dx = x^2 + C$$

Q8. Find the slope of $y = 3x + 7$

$$\text{Slope} = 3$$

Q9. Convert 45 degrees to radians.

$$45 \text{ degrees} = \pi/4 \text{ radians} = 0.7854 \text{ radians}$$

Q10. What is the square root of 144?

$$\sqrt{144} = 12$$